

Rolling a 12-word seed — the method

This sheet explains how to use the BIP-39 word table on the pages after the worksheet, for a **12-word seed**. The full procedure, including what to do with the words once you have them, is at bitcoinkeys.guide/roll-your-own-seed — read it before you begin. Do not work from this sheet alone.

What you need

One six-sided die, one coin, pen and paper. No computer, no phone.

How many words you roll

11 words. Your seed is 12 words long, and you roll all but the last one. **You do not roll word 12.** It carries a checksum that has to be calculated, so there is no entry for it in the table and no way to reach it with a die — your hardware wallet works it out, and the procedure on the site explains how for each device.

For each of your 11 words

1. **Throw the die 5 times. Reroll every 5 and 6** — they do not count and are not written down. Keep throwing until you have 5 results that are each 1, 2, 3 or 4. This is why no entry in the BIP-39 word table contains a 5 or a 6.
2. **Flip the coin once.** H is heads, T is tails. The coin is not optional: the die gives ten of the eleven bits a word needs and the coin gives the eleventh. Always taking the same side halves the randomness of every word you pick.
3. **Find your word.** Blocks are headed by your first 3 throws. Find that block **in the BIP-39 word table**, then read down to the line matching your remaining 2 throws and your flip.
4. **Write the word down, in order, by hand.** Never type it, photograph it, or read it aloud in earshot of anything with a microphone.

Two things worth knowing

The BIP-39 word table holds no secret. It is the standard word list in a fixed order, and anyone can regenerate it. Printing it, photographing it or keeping it on a phone is harmless. Your *throws* and the *words* they produce are the secret, and those never touch a screen.

This is the same method and the same table BitBox publishes for its own devices. We generated ours independently from the word list and checked all 2,048 entries against theirs: every one matches. Either table gives the same word for the same throws — but use one table for a whole seed, not a mix.

Worksheet — 12-word seed

The moment you write on this sheet it becomes your seed. Treat it exactly as you would the words themselves: never photograph it, never type it up, never leave it out. When your words are safely on their permanent backup and you have restored from that backup successfully, destroy this sheet — shred or burn it. Do not simply bin it.

WORD #	BLOCK (1st 3 rolls)			LINE (next 2)		FLIP (H/T)	WORD (BIP-39 table)
							<i>globe</i>
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12	Not rolled — your device calculates this one						

This sheet is for a 12-word seed: you roll 11 words and your device supplies word 12, which carries the checksum. A 24-word seed has its own sheet — **print only the one you are doing**. Full procedure: bitcoinkeys.guide/roll-your-own-seed

Rolling a 24-word seed — the method

This sheet explains how to use the BIP-39 word table on the pages after the worksheet, for a **24-word seed**. The full procedure, including what to do with the words once you have them, is at bitcoinkeys.guide/roll-your-own-seed — read it before you begin. Do not work from this sheet alone.

What you need

One six-sided die, one coin, pen and paper. No computer, no phone.

How many words you roll

23 words. Your seed is 24 words long, and you roll all but the last one. **You do not roll word 24.** It carries a checksum that has to be calculated, so there is no entry for it in the table and no way to reach it with a die — your hardware wallet works it out, and the procedure on the site explains how for each device.

For each of your 23 words

1. **Throw the die 5 times. Reroll every 5 and 6** — they do not count and are not written down. Keep throwing until you have 5 results that are each 1, 2, 3 or 4. This is why no entry in the BIP-39 word table contains a 5 or a 6.
2. **Flip the coin once.** H is heads, T is tails. The coin is not optional: the die gives ten of the eleven bits a word needs and the coin gives the eleventh. Always taking the same side halves the randomness of every word you pick.
3. **Find your word.** Blocks are headed by your first 3 throws. Find that block **in the BIP-39 word table**, then read down to the line matching your remaining 2 throws and your flip.
4. **Write the word down, in order, by hand.** Never type it, photograph it, or read it aloud in earshot of anything with a microphone.

Two things worth knowing

The BIP-39 word table holds no secret. It is the standard word list in a fixed order, and anyone can regenerate it. Printing it, photographing it or keeping it on a phone is harmless. Your *throws* and the *words* they produce are the secret, and those never touch a screen.

This is the same method and the same table BitBox publishes for its own devices. We generated ours independently from the word list and checked all 2,048 entries against theirs: every one matches. Either table gives the same word for the same throws — but use one table for a whole seed, not a mix.

Worksheet — 24-word seed

The moment you write on this sheet it becomes your seed. Treat it exactly as you would the words themselves: never photograph it, never type it up, never leave it out. When your words are safely on their permanent backup and you have restored from that backup successfully, destroy this sheet — shred or burn it. Do not simply bin it.

WORD #	BLOCK (1st 3 rolls)			LINE (next 2)		FLIP (H/T)	WORD (BIP-39 table)
							<i>globe</i>
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							
24	Not rolled — your device calculates this one						

This sheet is for a 24-word seed: you roll 23 words and your device supplies word 24, which carries the checksum. A 12-word seed has its own sheet — print only the one you are doing. Full procedure: bitcoinkeys.guide/roll-your-own-seed

11

111

11H abandon	31H acoustic
11T ability	31T acquire
12H able	32H across
12T about	32T act
13H above	33H action
13T absent	33T actor
14H absorb	34H actress
14T abstract	34T actual
21H absurd	41H adapt
21T abuse	41T add
22H access	42H addict
22T accident	42T address
23H account	43H adjust
23T accuse	43T admit
24H achieve	44H adult
24T acid	44T advance

112

11H advice	31H alcohol
11T aerobic	31T alert
12H affair	32H alien
12T afford	32T all
13H afraid	33H alley
13T again	33T allow
14H age	34H almost
14T agent	34T alone
21H agree	41H alpha
21T ahead	41T already
22H aim	42H also
22T air	42T alter
23H airport	43H always
23T aisle	43T amateur
24H alarm	44H amazing
24T album	44T among

113

11H amount	31H anxiety
11T amused	31T any
12H analyst	32H apart
12T anchor	32T apology
13H ancient	33H appear
13T anger	33T apple
14H angle	34H approve
14T angry	34T april
21H animal	41H arch
21T ankle	41T arctic
22H announce	42H area
22T annual	42T arena
23H another	43H argue
23T answer	43T arm
24H antenna	44H armed
24T antique	44T armor

114

11H army	31H asthma
11T around	31T athlete
12H arrange	32H atom
12T arrest	32T attack
13H arrive	33H attend
13T arrow	33T attitude
14H art	34H attract
14T artefact	34T auction
21H artist	41H audit
21T artwork	41T august
22H ask	42H aunt
22T aspect	42T author
23H assault	43H auto
23T asset	43T autumn
24H assist	44H average
24T assume	44T avocado

12

121

11H avoid	31H bamboo
11T awake	31T banana
12H aware	32H banner
12T away	32T bar
13H awesome	33H barely
13T awful	33T bargain
14H awkward	34H barrel
14T axis	34T base
21H baby	41H basic
21T bachelor	41T basket
22H bacon	42H battle
22T badge	42T beach
23H bag	43H bean
23T balance	43T beauty
24H balcony	44H because
24T ball	44T become

122

11H beef	31H bid
11T before	31T bike
12H begin	32H bind
12T behave	32T biology
13H behind	33H bird
13T believe	33T birth
14H below	34H bitter
14T belt	34T black
21H bench	41H blade
21T benefit	41T blame
22H best	42H blanket
22T betray	42T blast
23H better	43H bleak
23T between	43T bless
24H beyond	44H blind
24T bicycle	44T blood

123

11H blossom	31H borrow
11T blouse	31T boss
12H blue	32H bottom
12T blur	32T bounce
13H blush	33H box
13T board	33T boy
14H boat	34H bracket
14T body	34T brain
21H boil	41H brand
21T bomb	41T brass
22H bone	42H brave
22T bonus	42T bread
23H book	43H breeze
23T boost	43T brick
24H border	44H bridge
24T boring	44T brief

124

11H bright	31H bulk
11T bring	31T bullet
12H brisk	32H bundle
12T broccoli	32T bunker
13H broken	33H burden
13T bronze	33T burger
14H broom	34H burst
14T brother	34T bus
21H brown	41H business
21T brush	41T busy
22H bubble	42H butter
22T buddy	42T buyer
23H budget	43H buzz
23T buffalo	43T cabbage
24H build	44H cabin
24T bulb	44T cable

13

131

11H cactus	31H capital
11T cage	31T captain
12H cake	32H car
12T call	32T carbon
13H calm	33H card
13T camera	33T cargo
14H camp	34H carpet
14T can	34T carry
21H canal	41H cart
21T cancel	41T case
22H candy	42H cash
22T cannon	42T casino
23H canoe	43H castle
23T canvas	43T casual
24H canyon	44H cat
24T capable	44T catalog

132

11H catch	31H champion
11T category	31T change
12H cattle	32H chaos
12T caught	32T chapter
13H cause	33H charge
13T caution	33T chase
14H cave	34H chat
14T ceiling	34T cheap
21H celery	41H check
21T cement	41T cheese
22H census	42H chef
22T century	42T cherry
23H cereal	43H chest
23T certain	43T chicken
24H chair	44H chick
24T chalk	44T child

133

11H chimney	31H claw
11T choice	31T clay
12H choose	32H clean
12T chronic	32T clerk
13H chuckle	33H clever
13T chunk	33T click
14H churn	34H client
14T cigar	34T cliff
21H cinnamon	41H climb
21T circle	41T clinic
22H citizen	42H clip
22T city	42T clock
23H civil	43H clog
23T claim	43T close
24H clap	44H cloth
24T clarify	44T cloud

134

11H clown	31H come
11T club	31T comfort
12H clump	32H comic
12T cluster	32T common
13H clutch	33H company
13T coach	33T concert
14H coast	34H conduct
14T coconut	34T confirm
21H code	41H congress
21T coffee	41T connect
22H coil	42H consider
22T coin	42T control
23H collect	43H convince
23T color	43T cook
24H column	44H cool
24T combine	44T copper

14

141

11H copy	31H craft
11T coral	31T cram
12H core	32H crane
12T corn	32T crash
13H correct	33H crater
13T cost	33T crawl
14H cotton	34H crazy
14T couch	34T cream
21H country	41H credit
21T couple	41T creek
22H course	42H crew
22T cousin	42T cricket
23H cover	43H crime
23T coyote	43T crisp
24H crack	44H critic
24T cradle	44T crop

142

11H cross	31H current
11T crouch	31T curtain
12H crowd	32H curve
12T crucial	32T cushion
13H cruel	33H custom
13T cruise	33T cute
14H crumble	34H cycle
14T crunch	34T dad
21H crush	41H damage
21T cry	41T damp
22H crystal	42H dance
22T cube	42T danger
23H culture	43H daring
23T cup	43T dash
24H cupboard	44H daughter
24T curious	44T dawn

143

11H day	31H deliver
11T deal	31T demand
12H debate	32H demise
12T debris	32T denial
13H decade	33H dentist
13T december	33T deny
14H decide	34H depart
14T decline	34T depend
21H decorate	41H deposit
21T decrease	41T depth
22H deer	42H deputy
22T defense	42T derive
23H define	43H describe
23T defy	43T desert
24H degree	44H design
24T delay	44T desk

144

11H despair	31H dignity
11T destroy	31T dilemma
12H detail	32H dinner
12T detect	32T dinosaur
13H develop	33H direct
13T device	33T dirt
14H devote	34H disagree
14T diagram	34T discover
21H dial	41H disease
21T diamond	41T dish
22H diary	42H dismiss
22T dice	42T disorder
23H diesel	43H display
23T diet	43T distance
24H differ	44H divert
24T digital	44T divide

21

211		
11H divorce	31H dragon	
11T dizzy	31T drama	
12H doctor	32H drastic	
12T document	32T draw	
13H dog	33H dream	
13T doll	33T dress	
14H dolphin	34H drift	
14T domain	34T drill	
21H donate	41H drink	
21T donkey	41T drip	
22H donor	42H drive	
22T door	42T drop	
23H dose	43H drum	
23T double	43T dry	
24H dove	44H duck	
24T draft	44T dumb	

212		
11H dune	31H ecology	
11T during	31T economy	
12H dust	32H edge	
12T dutch	32T edit	
13H duty	33H educate	
13T dwarf	33T effort	
14H dynamic	34H egg	
14T eager	34T eight	
21H eagle	41H either	
21T early	41T elbow	
22H earn	42H elder	
22T earth	42T electric	
23H easily	43H elegant	
23T east	43T element	
24H easy	44H elephant	
24T echo	44T elevator	

213		
11H elite	31H energy	
11T else	31T enforce	
12H embark	32H engage	
12T embody	32T engine	
13H embrace	33H enhance	
13T emerge	33T enjoy	
14H emotion	34H enlist	
14T employ	34T enough	
21H empower	41H enrich	
21T empty	41T enroll	
22H enable	42H ensure	
22T enact	42T enter	
23H end	43H entire	
23T endless	43T entry	
24H endorse	44H envelope	
24T enemy	44T episode	

214		
11H equal	31H evoke	
11T equip	31T evolve	
12H era	32H exact	
12T erase	32T example	
13H erode	33H excess	
13T erosion	33T exchange	
14H error	34H excite	
14T erupt	34T exclude	
21H escape	41H excuse	
21T essay	41T execute	
22H essence	42H exercise	
22T estate	42T exhaust	
23H eternal	43H exhibit	
23T ethics	43T exile	
24H evidence	44H exist	
24T evil	44T exit	

22

221		
11H exotic	31H faith	
11T expand	31T fall	
12H expect	32H false	
12T expire	32T fame	
13H explain	33H family	
13T expose	33T famous	
14H express	34H fan	
14T extend	34T fancy	
21H extra	41H fantasy	
21T eye	41T farm	
22H eyebrow	42H fashion	
22T fabric	42T fat	
23H face	43H fatal	
23T faculty	43T father	
24H fade	44H fatigue	
24T faint	44T fault	

222		
11H favorite	31H figure	
11T feature	31T file	
12H february	32H film	
12T federal	32T filter	
13H fee	33H final	
13T feed	33T find	
14H feel	34H fine	
14T female	34T finger	
21H fence	41H finish	
21T festival	41T fire	
22H fetch	42H firm	
22T fever	42T first	
23H few	43H fiscal	
23T fiber	43T fish	
24H fiction	44H fit	
24T field	44T fitness	

223		
11H fix	31H foam	
11T flag	31T focus	
12H flame	32H fog	
12T flash	32T foil	
13H flat	33H fold	
13T flavor	33T follow	
14H flee	34H food	
14T flight	34T foot	
21H flip	41H force	
21T float	41T forest	
22H flock	42H forget	
22T floor	42T fork	
23H flower	43H fortune	
23T fluid	43T forum	
24H flush	44H forward	
24T fly	44T fossil	

224		
11H foster	31H fun	
11T found	31T funny	
12H fox	32H furnace	
12T fragile	32T fury	
13H frame	33H future	
13T frequent	33T gadget	
14H fresh	34H gain	
14T friend	34T galaxy	
21H fringe	41H gallery	
21T frog	41T game	
22H front	42H gap	
22T frost	42T garage	
23H frown	43H garbage	
23T frozen	43T garden	
24H fruit	44H garlic	
24T fuel	44T garment	

23

231		
11H gas	31H ginger	
11T gasp	31T giraffe	
12H gate	32H girl	
12T gather	32T give	
13H gauge	33H glad	
13T gaze	33T glance	
14H general	34H glare	
14T genius	34T glass	
21H genre	41H glide	
21T gentle	41T glimpse	
22H genuine	42H globe	
22T gesture	42T gloom	
23H ghost	43H glory	
23T giant	43T glove	
24H gift	44H glow	
24T giggle	44T glue	

232		
11H goat	31H gravity	
11T goddess	31T great	
12H gold	32H green	
12T good	32T grid	
13H goose	33H grief	
13T gorilla	33T grit	
14H gospel	34H grocery	
14T gossip	34T group	
21H govern	41H grow	
21T gown	41T grunt	
22H grab	42H guard	
22T grace	42T guess	
23H grain	43H guide	
23T grant	43T guilt	
24H grape	44H guitar	
24T grass	44T gun	

233		
11H gym	31H head	
11T habit	31T health	
12H hair	32H heart	
12T half	32T heavy	
13H hammer	33H hedgehog	
13T hamster	33T height	
14H hand	34H hello	
14T happy	34T helmet	
21H harbor	41H help	
21T hard	41T hen	
22H harsh	42H hero	
22T harvest	42T hidden	
23H hat	43H high	
23T have	43T hill	
24H hawk	44H hint	
24T hazard	44T hip	

234		
11H hire	31H host	
11T history	31T hotel	
12H hobby	32H hour	
12T hockey	32T hover	
13H hold	33H hub	
13T hole	33T huge	
14H holiday	34H human	
14T hollow	34T humble	
21H home	41H humor	
21T honey	41T hundred	
22H hood	42H hungry	
22T hope	42T hunt	
23H horn	43H hurdle	
23T horror	43T hurry	
24H horse	44H hurt	
24T hospital	44T husband	

24

241		
11H hybrid	31H improve	
11T ice	31T impulse	
12H icon	32H inch	
12T idea	32T include	
13H identify	33H income	
13T idle	33T increase	
14H ignore	34H index	
14T ill	34T indicate	
21H illegal	41H indoor	
21T illness	41T industry	
22H image	42H infant	
22T imitate	42T inflict	
23H immense	43H inform	
23T immune	43T inhale	
24H impact	44H inherit	
24T impose	44T initial	

242		
11H inject	31H invite	
11T injury	31T involve	
12H inmate	32H iron	
12T inner	32T island	
13H innocent	33H isolate	
13T input	33T issue	
14H inquiry	34H item	
14T insane	34T ivory	
21H insect	41H jacket	
21T inside	41T jaguar	
22H inspire	42H jar	
22T install	42T jazz	
23H intact	43H jealous	
23T interest	43T jeans	
24H into	44H jelly	
24T invest	44T jewel	

243		
11H job	31H key	
11T join	31T kick	
12H joke	32H kid	
12T journey	32T kidney	
13H joy	33H kind	
13T judge	33T kingdom	
14H juice	34H kiss	
14T jump	34T kit	
21H jungle	41H kitchen	
21T junior	41T kite	
22H junk	42H kitten	
22T just	42T kiwi	
23H kangaroo	43H knee	
23T keen	43T knife	
24H keep	44H knock	
24T ketchup	44T know	

244		
11H lab	31H lawn	
11T label	31T lawsuit	
12H labor	32H layer	
12T ladder	32T lazy	
13H lady	33H leader	
13T lake	33T leaf	
14H lamp	34H learn	
14T language	34T leave	
21H laptop	41H lecture	
21T large	41T left	
22H later	42H leg	
22T latin	42T legal	
23H laugh	43H legend	
23T laundry	43T leisure	
24H lava	44H lemon	
24T law	44T lend	

31

311

11H length	31H link
11T lens	31T lion
12H leopard	32H liquid
12T lesson	32T list
13H letter	33H little
13T level	33T live
14H liar	34H lizard
14T liberty	34T load
21H library	41H loan
21T license	41T lobster
22H life	42H local
22T lift	42T lock
23H light	43H logic
23T like	43T lonely
24H limb	44H long
24T limit	44T loop

312

11H lottery	31H maid
11T loud	31T mail
12H lounge	32H main
12T love	32T major
13H loyal	33H make
13T lucky	33T mammal
14H luggage	34H man
14T lumber	34T manage
21H lunar	41H mandate
21T lunch	41T mango
22H luxury	42H mansion
22T lyrics	42T manual
23H machine	43H maple
23T mad	43T marble
24H magic	44H march
24T magnet	44T margin

313

11H marine	31H meat
11T market	31T mechanic
12H marriage	32H medal
12T mask	32T media
13H mass	33H melody
13T master	33T melt
14H match	34H member
14T material	34T memory
21H math	41H mention
21T matrix	41T menu
22H matter	42H mercy
22T maximum	42T merge
23H maze	43H merit
23T meadow	43T merry
24H mean	44H mesh
24T measure	44T message

314

11H metal	31H mix
11T method	31T mixed
12H middle	32H mixture
12T midnight	32T mobile
13H milk	33H model
13T million	33T modify
14H mimic	34H mom
14T mind	34T moment
21H minimum	41H monitor
21T minor	41T monkey
22H minute	42H monster
22T miracle	42T month
23H mirror	43H moon
23T misery	43T moral
24H miss	44H more
24T mistake	44T morning

32

321

11H mosquito	31H must
11T mother	31T mutual
12H motion	32H myself
12T motor	32T mystery
13H mountain	33H myth
13T mouse	33T naive
14H move	34H name
14T movie	34T napkin
21H much	41H narrow
21T muffin	41T nasty
22H mule	42H nation
22T multiply	42T nature
23H muscle	43H near
23T museum	43T neck
24H mushroom	44H need
24T music	44T negative

322

11H neglect	31H noodle
11T neither	31T normal
12H nephew	32H north
12T nerve	32T nose
13H nest	33H notable
13T net	33T note
14H network	34H nothing
14T neutral	34T notice
21H never	41H novel
21T news	41T now
22H next	42H nuclear
22T nice	42T number
23H night	43H nurse
23T noble	43T nut
24H noise	44H oak
24T nominee	44T obey

323

11H object	31H old
11T oblige	31T olive
12H obscure	32H olympic
12T observe	32T omit
13H obtain	33H once
13T obvious	33T one
14H occur	34H onion
14T ocean	34T online
21H october	41H only
21T odor	41T open
22H off	42H opera
22T offer	42T opinion
23H office	43H oppose
23T often	43T option
24H oil	44H orange
24T okay	44T orbit

324

11H orchard	31H own
11T order	31T owner
12H ordinary	32H oxygen
12T organ	32T oyster
13H orient	33H ozone
13T original	33T pact
14H orphan	34H paddle
14T ostrich	34T page
21H other	41H pair
21T outdoor	41T palace
22H outer	42H palm
22T output	42T panda
23H outside	43H panel
23T oval	43T panic
24H oven	44H panther
24T over	44T paper

33

331

11H parade	31H pear
11T parent	31T peasant
12H park	32H pelican
12T parrot	32T pen
13H party	33H penalty
13T pass	33T pencil
14H patch	34H people
14T path	34T pepper
21H patient	41H perfect
21T patrol	41T permit
22H pattern	42H person
22T pause	42T pet
23H pave	43H phone
23T payment	43T photo
24H peace	44H phrase
24T peanut	44T physical

332

11H piano	31H plastic
11T picnic	31T plate
12H picture	32H play
12T piece	32T please
13H pig	33H pledge
13T pigeon	33T pluck
14H pill	34H plug
14T pilot	34T plunge
21H pink	41H poem
21T pioneer	41T poet
22H pipe	42H point
22T pistol	42T polar
23H pitch	43H pole
23T pizza	43T police
24H place	44H pond
24T planet	44T pony

333

11H pool	31H present
11T popular	31T pretty
12H portion	32H prevent
12T position	32T price
13H possible	33H pride
13T post	33T primary
14H potato	34H print
14T pottery	34T priority
21H poverty	41H prison
21T powder	41T private
22H power	42H prize
22T practice	42T problem
23H praise	43H process
23T predict	43T produce
24H prefer	44H profit
24T prepare	44T program

334

11H project	31H puppy
11T promote	31T purchase
12H proof	32H purity
12T property	32T purpose
13H prosper	33H purse
13T protect	33T push
14H proud	34H put
14T provide	34T puzzle
21H public	41H pyramid
21T pudding	41T quality
22H pull	42H quantum
22T pulp	42T quarter
23H pulse	43H question
23T pumpkin	43T quick
24H punch	44H quit
24T pupil	44T quiz

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341

11H quote	31H rare
11T rabbit	31T rate
12H raccoon	32H rather
12T race	32T raven
13H rack	33H raw
13T radar	33T razor
14H radio	34H ready
14T rail	34T real
21H rain	41H reason
21T raise	41T rebel
22H rally	42H rebuild
22T ramp	42T recall
23H ranch	43H receive
23T random	43T recipe
24H range	44H record
24T rapid	44T recycle

342

11H reduce	31H render
11T reflect	31T renew
12H reform	32H rent
12T refuse	32T reopen
13H region	33H repair
13T regret	33T repeat
14H regular	34H replace
14T reject	34T report
21H relax	41H require
21T release	41T rescue
22H relief	42H resemble
22T rely	42T resist
23H remain	43H resource
23T remember	43T response
24H remind	44H result
24T remove	44T retire

343

11H retreat	31H ring
11T return	31T riot
12H reunion	32H ripple
12T reveal	32T risk
13H review	33H ritual
13T reward	33T rival
14H rhythm	34H river
14T rib	34T road
21H ribbon	41H roast
21T rice	41T robot
22H rich	42H robust
22T ride	42T rocket
23H ridge	43H romance
23T rifle	43T roof
24H right	44H rookie
24T rigid	44T room

344

11H rose	31H safe
11T rotate	31T sail
12H rough	32H salad
12T round	32T salmon
13H route	33H salon
13T royal	33T salt
14H rubber	34H salute
14T rude	34T same
21H rug	41H sample
21T rule	41T sand
22H run	42H satisfy
22T runway	42T satoshi
23H rural	43H sauce
23T sad	43T sausage
24H saddle	44H save
24T sadness	44T say

41

411		412		413		414	
11H scale	31H search	11H series	31H ship	11H siege	31H skate	11H slot	31H sock
11T scan	31T season	11T service	31T shiver	11T sight	31T sketch	11T slow	31T soda
12H scare	32H seat	12H session	32H shock	12H sign	32H ski	12H slush	32H soft
12T scatter	32T second	12T settle	32T shoe	12T silent	32T skill	12T small	32T solar
13H scene	33H secret	13H setup	33H shoot	13H silk	33H skin	13H smart	33H soldier
13T scheme	33T section	13T seven	33T shop	13T silly	33T skirt	13T smile	33T solid
14H school	34H security	14H shadow	34H short	14H silver	34H skull	14H smoke	34H solution
14T science	34T seed	14T shaft	34T shoulder	14T similar	34T slab	14T smooth	34T solve
21H scissors	41H seek	21H shallow	41H shove	21H simple	41H slam	21H snack	41H someone
21T scorpion	41T segment	21T share	41T shrimp	21T since	41T sleep	21T snake	41T song
22H scout	42H select	22H shed	42H shrug	22H sing	42H slender	22H snap	42H soon
22T scrap	42T sell	22T shell	42T shuffle	22T siren	42T slice	22T sniff	42T sorry
23H screen	43H seminar	23H sheriff	43H shy	23H sister	43H slide	23H snow	43H sort
23T script	43T senior	23T shield	43T sibling	23T situate	43T slight	23T soap	43T soul
24H scrub	44H sense	24H shift	44H sick	24H six	44H slim	24H soccer	44H sound
24T sea	44T sentence	24T shine	44T side	24T size	44T slogan	24T social	44T soup

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421		422		423		424	
11H source	31H spirit	11H staff	31H stock	11H submit	31H sure	11H swing	31H task
11T south	31T split	11T stage	31T stomach	11T subway	31T surface	11T switch	31T taste
12H space	32H spoil	12H stairs	32H stone	12H success	32H surge	12H sword	32H tattoo
12T spare	32T sponsor	12T stamp	32T stool	12T such	32T surprise	12T symbol	32T taxi
13H spatial	33H spoon	13H stand	33H story	13H sudden	33H surround	13H symptom	33H teach
13T spawn	33T sport	13T start	33T stove	13T suffer	33T survey	13T syrup	33T team
14H speak	34H spot	14H state	34H strategy	14H sugar	34H suspect	14H system	34H tell
14T special	34T spray	14T stay	34T street	14T suggest	34T sustain	14T table	34T ten
21H speed	41H spread	21H steak	41H strike	21H suit	41H swallow	21H tackle	41H tenant
21T spell	41T spring	21T steel	41T strong	21T summer	41T swamp	21T tag	41T tennis
22H spend	42H spy	22H stem	42H struggle	22H sun	42H swap	22H tail	42H tent
22T sphere	42T square	22T step	42T student	22T sunny	42T swarm	22T talent	42T term
23H spice	43H squeeze	23H stereo	43H stuff	23H sunset	43H swear	23H talk	43H test
23T spider	43T squirrel	23T stick	43T stumble	23T super	43T sweet	23T tank	43T text
24H spike	44H stable	24H still	44H style	24H supply	44H swift	24H tape	44H thank
24T spin	44T stadium	24T sting	44T subject	24T supreme	44T swim	24T target	44T that

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431		432		433		434	
11H theme	31H tilt	11H tomato	31H toward	11H trend	31H tube	11H ugly	31H unlock
11T then	31T timber	11T tomorrow	31T tower	11T trial	31T tuition	11T umbrella	31T until
12H theory	32H time	12H tone	32H town	12H tribe	32H tumble	12H unable	32H unusual
12T there	32T tiny	12T tongue	32T toy	12T trick	32T tuna	12T unaware	32T unveil
13H they	33H tip	13H tonight	33H track	13H trigger	33H tunnel	13H uncle	33H update
13T thing	33T tired	13T tool	33T trade	13T trim	33T turkey	13T uncover	33T upgrade
14H this	34H tissue	14H tooth	34H traffic	14H trip	34H turn	14H under	34H uphold
14T thought	34T title	14T top	34T tragic	14T trophy	34T turtle	14T undo	34T upon
21H three	41H toast	21H topic	41H train	21H trouble	41H twelve	21H unfair	41H upper
21T thrive	41T tobacco	21T topple	41T transfer	21T truck	41T twenty	21T unfold	41T upset
22H throw	42H today	22H torch	42H trap	22H true	42H twice	22H unhappy	42H urban
22T thumb	42T toddler	22T tornado	42T trash	22T truly	42T twin	22T uniform	42T urge
23H thunder	43H toe	23H tortoise	43H travel	23H trumpet	43H twist	23H unique	43H usage
23T ticket	43T together	23T toss	43T tray	23T trust	43T two	23T unit	43T use
24H tide	44H toilet	24H total	44H treat	24H truth	44H type	24H universe	44H used
24T tiger	44T token	24T tourist	44T tree	24T try	44T typical	24T unknown	44T useful

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441		442		443		444	
11H useless	31H velvet	11H village	31H voyage	11H way	31H wheel	11H winter	31H wrap
11T usual	31T vendor	11T vintage	31T wage	11T wealth	31T when	11T wire	31T wreck
12H utility	32H venture	12H violin	32H wagon	12H weapon	32H where	12H wisdom	32H wrestle
12T vacant	32T venue	12T virtual	32T wait	12T wear	32T whip	12T wise	32T wrist
13H vacuum	33H verb	13H virus	33H walk	13H weasel	33H whisper	13H wish	33H write
13T vague	33T verify	13T visa	33T wall	13T weather	33T wide	13T witness	33T wrong
14H valid	34H version	14H visit	34H walnut	14H web	34H width	14H wolf	34H yard
14T valley	34T very	14T visual	34T want	14T wedding	34T wife	14T woman	34T year
21H valve	41H vessel	21H vital	41H warfare	21H weekend	41H wild	21H wonder	41H yellow
21T van	41T veteran	21T vivid	41T warm	21T weird	41T will	21T wood	41T you
22H vanish	42H viable	22H vocal	42H warrior	22H welcome	42H win	22H wool	42H young
22T vapor	42T vibrant	22T voice	42T wash	22T west	42T window	22T word	42T youth
23H various	43H vicious	23H void	43H wasp	23H wet	43H wine	23H work	43H zebra
23T vast	43T victory	23T volcano	43T waste	23T whale	43T wing	23T world	43T zero
24H vault	44H video	24H volume	44H water	24H what	44H wink	24H worry	44H zone
24T vehicle	44T view	24T vote	44T wave	24T wheat	44T winner	24T worth	44T zoo